

BATCH : 2019-2023

FOR : Internship

COMPANY : SAP Labs, Bangalore

ABOUT THE COMPANY : SAP SE is a German multinational software corporation that develops enterprise software to manage business operations and customer relations.

NUMBER OF ROUNDS : 4

1. Online Test
2. 2 Technical Rounds
3. 1 HR Round

ONLINE TEST :

This round consists of two coding questions with a duration of 45 min .

1.

Input : An unsorted array and an integer k. ($k \geq 1$)

Output : Print the kth smallest element in the unsorted array.

[K'th Smallest / Largest Element in Unsorted Array](#)

2.

Input : An integer array of length n with elements 0 to n-1.

Output : Print an array where the elements have to be rearranged, where the elements are taken as indices and the indices as elements .

Example :

Input : [3, 0, 1, 2]

Output : [1, 2, 3, 0]

Let a be the input array .

$a[0] = 3$; $a[1] = 0$; $a[2] = 1$; $a[3] = 2$

Output should be :

$a[3] = 0$; $a[0] = 1$; $a[1] = 2$; $a[2] = 3$

1st TECHNICAL ROUND : 35 - 40 min approx.

- I was asked to introduce myself and then asked to explain about any one of my projects. I explained a project related to Database Management System, after which I was asked to write 2 SQL queries based on the relations used in my project. I was asked to share my screen with a Notepad open for writing the queries.
First query was based on aggregate function.
Second query was based on Joins.
- Then, I was asked to explain the solution to the following problem.

Removing duplicates in an unsorted array.

(Tip - Start from Naive approach , then proceed to optimised solution)

[Remove duplicates from an unsorted array](#)

I was asked to type the code for the naive approach in the same notepad.

- After which I was asked questions on various CS concepts. I chose to explain in CPP wherever necessary.
 - How is exception handling done in CPP ?
 - Explain the difference between array and linked list.
 - Difference between Unique and Distinct in SQL.
 - Explain various constraints in SQL.
- Finally, I was asked to provide a solution for a puzzle. The puzzle is as follows :
Given a 5L jar and a 3L jar and an infinite supply of water, how will you measure 4L ?
(General Tip : Do not lose hope when you do not know the entire solution. It is our approach to the given problem that matters)

2nd TECHNICAL ROUND : 45 - 50 min approx.

- I was asked to introduce myself and was asked about my interview experience in the first round.
- I was asked to explain a project other than the one asked in the previous round.
- Then, I was asked to explain the concept of abstraction in OOPS with an example. I was asked to type the code in a notepad.
- After which I was asked a few questions in CS concepts :
 - Explain the types of indexing in DBMS.
 - What is the data structure used in LRU Cache ?
 - Explain the Threading concept in OS and the functions used.
 - What is Macro ?
 - In Python, which data type is mutable and which is immutable ? And why is tuple immutable ?
[Why is tuple immutable in Python](#)
(I was asked this question because I had mentioned Python in my resume)
 - Explain the difference between vector and list in STL.
- I was asked to explain call by value and call by reference where I had to type an example code in the same notepad.
- Then, I was asked to type the code for implementing a queue using 2 stacks in the Notepad. Since there was no explicit mention to code the stack data structure from scratch , I used stacks from STL in CPP.
[Implement queue using 2 stacks](#)
Then I was asked to explain the time complexity of various operations in the queue implemented.
- I was asked to solve a puzzle . The puzzle is as follows :
There are 3 ants in the 3 corners of an equilateral triangle . If the ants start to move at the same speed, what is the probability that the ants will never meet ?

- Finally, they asked me if I had any questions. I asked a couple of questions.

HR ROUND : 15 min

- I was asked to introduce myself and then explain how I secured a seat in CEG.
 - Then I was asked to put forth the questions I had.
- The interview was completed.

KEY TAKEAWAYS FROM THE INTERVIEW PROCESS :

- Know your resume well.
- Be strong with the CS concepts.
- Puzzles are compulsory when it comes to SAP. Try to go through the puzzles asked in SAP.
- Refer geeksforgeeks archives for a particular company before sitting for an online test or for the interview process for that company.
- Do not lose hope when you aren't able to solve a question fully.

TOPICS TO BE FOCUSED :

- OOPS
- Database Management Systems
- Data Structures and Algorithms
- OS
- Whatever you have put in your resume

BEST WISHES !!

Vasudha. E